

# Rendu: A Physics-Grounded Simulation Platform for Spatial Acoustic Learning

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**Abstract.** We present Rendu, a physics-grounded renderer for novel-view acoustic synthesis (NVAS). NVAS is limited by the availability of controllable, physically interpretable acoustic data: real recordings are sparse, while mainstream geometric simulators do not represent the low-frequency room modes that govern spatial variation below the Schroeder frequency. Rendu combines GPU finite-difference time-domain wave simulation with accelerated geometric ray tracing, and supports frequency-dependent boundaries, differentiable wall-admittance estimation, and export of decomposable acoustic ground truth. On the measured RAF Empty Room, Rendu achieves a 12.7% held-out RT60 error and a 1.9 dB energy-decay-curve error. Under the same set-up, it recovers low-frequency modal structure closer to RAF measurements than SoundSpaces. Building on this renderer, we propose a controllable dataset and held-out novel-receiver benchmark with material-aware annotations and hard cases involving occlusion, diffraction, coupled spaces, and dynamic trajectories, supporting editable, generative, and embodied spatial-audio models.

**Keywords:** NVAS · Spatial intelligence

## 1 Introduction

Novel-view synthesis has driven rapid development in visual spatial intelligence, but for applications such as immersive AR/VR, interactive embodied AI, and digital twins, sound reconstruction is equally crucial. Novel-view acoustic synthesis (NVAS) [1,2,5] has been proposed to modelling environment acoustic property with impulse response (IR/RIR) or predicting binaural response at unobserved poses. Unlike visual NVS, however, NVAS lacks a mature data and benchmark ecosystem. Real measured fields such as RAF [4] are scarce. More training and evaluation rely on synthetic acoustic simulators, such as SoundSpaces [3].

Such simulator uses geometric acoustics, which is efficient, scalable, and works well for high-frequency components. However, for low-frequency components, it cannot fully represent patterns such as standing-wave room modes, which dominate below the Schroeder frequency. Acoustic responses may vary sharply over short spatial distances. Existing synthetic datasets also provide limited material variation or rarely expose decomposable physical targets such as direct/early/late components, per-surface contributions, or frequency-dependent absorption.

**Table 1:** Validation of Rendu’s individual components against analytical solutions or measured RAF room.

| Component            | Reference                    | Error                               |
|----------------------|------------------------------|-------------------------------------|
| FDTD wave kernel     | Analytic shoebox modes       | 0.26%                               |
| Freq.-dep. boundary  | Image-source method          | $R(\theta) < 4\%$ ; $T_{30} < 10\%$ |
| Geometric ray tracer | Eyring theory                | 3.9%                                |
| Wave-ray crossover   | Wave/ray overlap band        | 2.8%                                |
| Real room (RAF)      | Measured RIRs, held-out RT60 | 12.7%; EDC 1.9 dB                   |

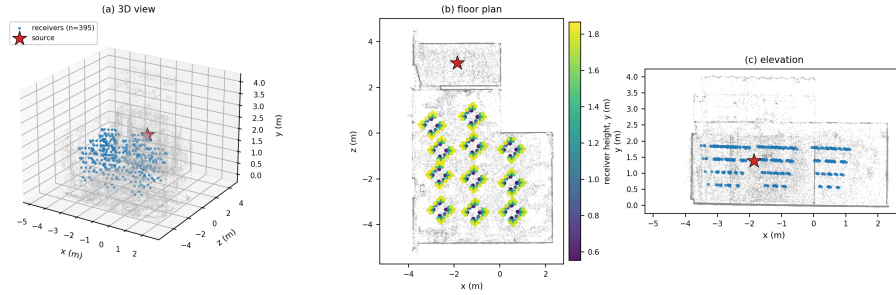
We thus introduce Rendu, a physics-grounded synthetic renderer that addresses these limitations.

## 2 The Rendu Renderer

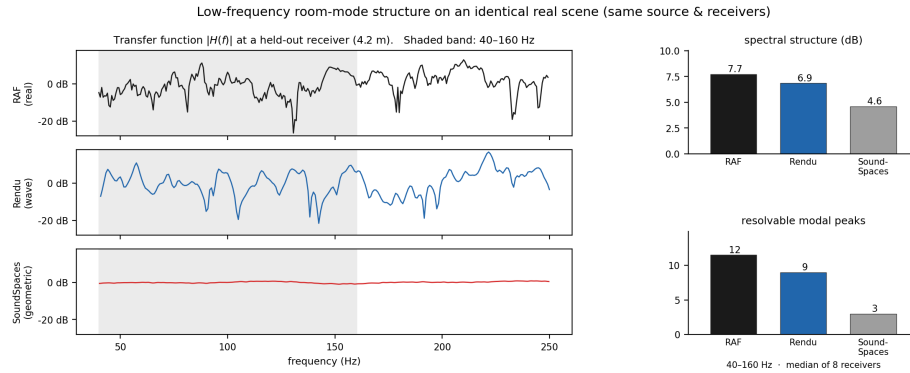
*Design.* Rendu combines a GPU-based finite-difference time-domain (FDTD) solver for low-frequency wave propagation with a geometric ray tracer for high-frequency sound transport. The system supports both synthetic scenes and reconstructed real environments, with a cross-platform PyTorch implementation and an efficient OptiX implementation. The FDTD solver uses a differentiable, frequency-dependent locally reacting impedance boundary, enabling wall-admittance estimation in real rooms when material properties are unknown. A crossover filter combines the wave- and ray-based responses into a full-band room impulse response (RIR). Rendu also exports detailed acoustic ground truth, including RT60, EDT, C50, DRR, and direct/early/late decomposition.

*Validation.* Table 1 summarizes validation of the core components against analytic solutions. We further reconstruct and calibrate the RAF Empty Room (Fig. 1) from official-provided geometry, then compare simulated RIRs with real recordings at held-out receiver positions. On this real-room evaluation, Rendu achieves a reverberation-time error of 12.7% and an energy-decay-curve (EDC) error of 1.9 dB, indicating that it captures the measured room’s overall reverberation behavior beyond controlled synthetic settings.

*Comparison.* A key advantage of Rendu is its ability to model low-frequency room modes that are absent from mainstream geometric acoustic simulations. On the same RAF scene, source position, and receiver set, we compare the low-frequency transfer function  $|H(f)|$  from real RAF measurements, Rendu’s wave solver, and SoundSpaces’ RLR audio propagation engine running on the same mesh. As shown in Fig. 2, below the Schroeder frequency, the real field and Rendu both exhibit dense modal structure, with median spectral variation of 7.7 dB and 6.9 dB, respectively, and 12 and 9 resolvable modal peaks in the 40–160 Hz band. In contrast, SoundSpaces produces a substantially smoother response, with 4.6 dB spectral variation and only 3 modal peaks (median over 8 receivers; the left panel shows a representative one).



**Fig. 1:** RAF Empty Room calibration setup: reconstructed geometry, loudspeaker source (red star), and dense multi-height receiver field.

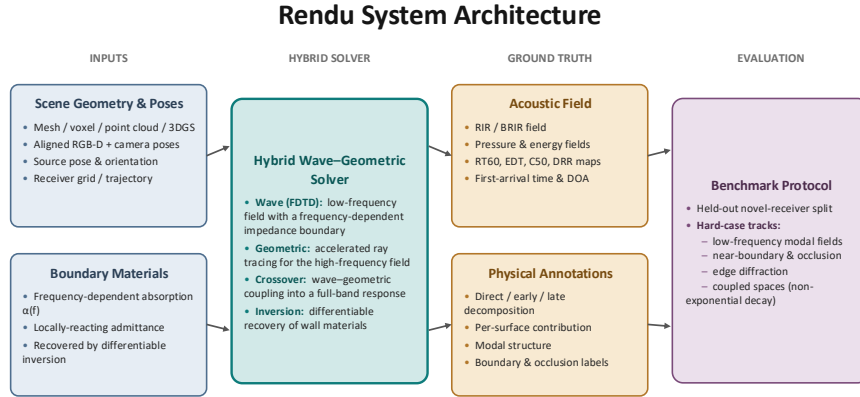


**Fig. 2:** Low-frequency modal structure on the same RAF scene. RAF and Rendu show dense modal variation, whereas SoundSpaces is nearly flat. Statistics over 40–160 Hz confirm  $\text{RAF} \approx \text{Rendu} \gg \text{SoundSpaces}$  in modal density and strength.

### 3 Dataset and Benchmark

*Dataset.* Rendu is designed to provide structured data for future model development beyond static RIR rendering (Fig. 3). It will support moving sound sources and listener trajectories with time-varying acoustic responses, together with decomposable ground truth such as direct/early/late separation, per-surface contributions, and frequency-dependent absorption. The dataset will further include aligned semantic and material labels. These annotations can support research on animated, editable, feed-forward novel-view acoustics, and embodied spatial-audio models.

*Benchmark.* As a benchmark, Rendu by default adopts a held-out novel-receiver protocol for evaluating spatial generalization at unseen listener positions. It will include targeted hard cases that expose failure modes not captured by standard geometric simulation, including occlusion, edge diffraction, coupled spaces,



**Fig. 3:** Rendu pipeline from scene inputs to physics-grounded acoustic fields and held-out novel-receiver evaluation.

mixed materials, and acoustically challenging geometries. Instead of reporting an aggregate audio error, the benchmark will use detailed physical annotations to identify whether a model fails because of modal structure, material response, reflection, visibility, or long-range reverberation. This enables more diagnostic evaluation and provides concrete grounds for improving future NVAS models.

*Future.* We will extend Rendu with an Unreal Engine front end for interactive acoustic authoring and visualization. We will investigate Adaptive Rectangular Decomposition (ARD) for fast preview, the Discontinuous Galerkin Finite Element Method (DG-FEM) for high-fidelity offline baking, and neural-accelerated propagation for scalable acoustic-field generation.

## References

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